Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
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Recitation 12 Solutions

Problem 12.1

Suppose we want to build a classifier that labels email as spam or not spam based on whether certain words are present in the email, based on a somewhat simplified probabilistic model. For a given email, let indicator random variable C be equal to 1 if the email is spam and 0 if the email is not spam, and $s \triangleq p_C(1)$. We have a dictionary of n words and use an indicator random variable Y_i to capture whether word i from the dictionary appears in the email ($Y_i = 1$) or not ($Y_i = 0$). We assume that $Y_1, \ldots, Y_n$ are conditionally independent given C[1], and $p_i \triangleq p_{Y_i|C}(1|1)$, $q_i \triangleq p_{Y_i|C}(1|0)$, $i = 1, \ldots, n$. Assume that parameters $s, p_1, \ldots, p_n, q_1, \ldots, q_n$ are unknown. To build the classifier, we will estimate these parameters from a training set of k emails. Email j in the training set has a label $c^{(j)}$, which tells us whether the email is spam or not. For email j, we also generate n features $y^{(j)} = \left(y_1^{(j)}, y_2^{(j)}, \ldots, y_n^{(j)}\right)$ where $y_i^{(j)}$ is the observed value of Y_i for this email. We shall assume that different emails are generated i.i.d.from the same distribution.

(a) First, we estimate the parameters $\theta = (s, p_1, p_2, \ldots, p_n, q_1, q_2, \ldots, q_n)$.

(i) As a warm-up before doing any calculations, what is a reasonable way to estimate p_i and q_i?

Solution: Obviously, answers will vary. You should notice that you can simply count the number of times each feature shows up in the spam emails, and divide by the total number of spam emails. (And do the same for the non-spam emails.)

(ii) Determine functions $f, g_1, \ldots, g_n, h_1, \ldots, h_n$ such that

$$\ell\left(\theta; c^{(1,\ldots,k)}, y_{1,\ldots,n}^{(1,\ldots,k)}\right) \triangleq \log\left(\prod_{j=1}^{k} p_{C,Y_1,\ldots,Y_n}(c^{(j)}, y_1^{(j)}, \ldots, y_n^{(j)}; \theta)\right)$$
$$= f(s) + \sum_{i=1}^{n} g_i(p_i) + \sum_{i=1}^{n} h_i(q_i).$$

Your answer may depend on the training data $\left\{c^{(1,\ldots,k)}, y_{1,\ldots,n}^{(1,\ldots,k)}\right\}$.

[1]This assumption of the model is not actually true for emails since certain words may be more likely to co-occur. Also, the model does not account for the ordering of the words or whether a word occurs multiple times.

Solution: We have

$$
\begin{aligned}
&\ell\left(\theta; c^{(1,\ldots,k)}, y_{1,\ldots,n}^{(1,\ldots,k)}\right) \\
&\quad \triangleq \log\left(\prod_{j=1}^{k} p_{C,Y_1,\ldots,Y_n}(c^{(j)}, y_1^{(j)}, \ldots, y_n^{(j)}; \theta)\right) \\
&\quad = \sum_{j=1}^{k} \log p_C(c^{(j)}; \theta) \sum_{i=1}^{n} \log p_{Y_i|X}(y_i^{(j)}|c^{(j)}; \theta) \qquad (1) \\
&\quad = \sum_{j=1}^{k} \Big[c^{(j)} \log s + (1-c^{(j)}) \log(1-s) \\
&\qquad + c^{(j)} \sum_{i=1}^{n} \left[y_i^{(j)} \log p_i + (1-y_i^{(j)}) \log(1-p_i)\right] \\
&\qquad + (1-c^{(j)}) \sum_{i=1}^{n} \left[y_i^{(j)} \log q_i + (1-y_i^{(j)}) \log(1-q_i)\right]\Big] \qquad (2) \\
&\quad = \log s \left(\sum_{j=1}^{k} c^{(j)}\right) + \log(1-s) \left(\sum_{j=1}^{k} 1 - c^{(j)}\right) \\
&\qquad + \sum_{i=1}^{n} \log p_i \left(\sum_{j=1}^{k} c^{(j)} y_i^{(j)}\right) + \log(1-p_i) \left(\sum_{j=1}^{k} c^{(j)} (1-y_i^{(j)})\right) \\
&\qquad + \sum_{i=1}^{n} \log q_i \left(\sum_{j=1}^{k} (1-c^{(j)}) y_i^{(j)}\right) + \log(1-q_i) \left(\sum_{j=1}^{k} (1-c^{(j)}) (1-y_i^{(j)})\right), \qquad (3)
\end{aligned}
$$

where to obtain (1) we have used conditional independence of $Y_1, \ldots, Y_n$ given C, to obtain (2) we substituted the Bernoulli distribution into the expression, and to obtain (3) we have rearranged the summations. Setting

$$
\begin{aligned}
f(s) &= \log s \left(\sum_{j=1}^{k} c^{(j)}\right) + \log(1-s) \left(\sum_{j=1}^{k} 1 - c^{(j)}\right) \\
g_i(p_i) &= \log p_i \left(\sum_{j=1}^{k} c^{(j)} y_i^{(j)}\right) + \log(1-p_i) \left(\sum_{j=1}^{k} c^{(j)} (1-y_i^{(j)})\right), \quad i = 1, \ldots, n \\
h_i(q_i) &= \log q_i \left(\sum_{j=1}^{k} (1-c^{(j)}) y_i^{(j)}\right) + \log(1-q_i) \left(\sum_{j=1}^{k} (1-c^{(j)}) (1-y_i^{(j)})\right), \quad i = 1, \ldots, n
\end{aligned}
$$

we obtain the desired factorization.

(iii) Determine the ML estimates $\hat{s}, \hat{p}_1, \ldots, \hat{p}_n, \hat{q}_1, \ldots, \hat{q}_n$ of the model parameters.

Solution: Since the log-likelihood function factors as we saw in part (ii), we can estimate each parameter separately by differentiation, setting the derivative to zero, and verifying that indeed the estimate maximizes the function. Specifically, for s, we proceed as follows:

$$\begin{aligned}\frac{\partial f(s)}{\partial s} &= \frac{\sum_{j=1}^{k} c^{(j)}}{s} - \frac{\sum_{j=1}^{k} 1 - c^{(j)}}{1-s} \\ &= \frac{-ks + \sum_{j=1}^{k} c^{(j)}}{s(1-s)},\end{aligned}$$

which is equal to zero for

$$\hat{s} = \frac{1}{k} \sum_{j=1}^{k} c^{(j)}.$$

It is straightforward to verify that $\hat{s}$ is indeed a maximizer.

Similarly, we obtain

$$\hat{p}_i = \frac{1}{\sum_{j=1}^{k} c^{(j)}} \sum_{j=1}^{k} c^{(j)}\, y_i^{(j)}, \quad i = 1, \ldots, n,$$

$$\hat{q}_i = \frac{1}{\sum_{j=1}^{k} 1 - c^{(j)}} \sum_{j=1}^{k} (1 - c^{(j)})\, y_i^{(j)}, \quad i = 1, \ldots, n.$$

Compare these estimates with your proposed plan for estimating the model parameters in part (a)(i).

(b) Once we estimate the model parameters, we can treat them as fixed and classify new emails as spam or not spam.

(i) Suppose the new email has features $Y_1 = y_1, \ldots, Y_n = y_n$. Determine the probability that $C = 1$ (the email is spam).

Solution: Using Bayes' rule,

$$\begin{aligned}&p_{C|Y_1,\ldots,Y_n}(1|y_1,\ldots,y_n;\hat{\theta}) \\ &= \frac{p_C(1;\hat{\theta})\, p_{Y_1,\ldots,Y_n|C}(y_1,\ldots,y_n|1;\hat{\theta})}{p_{Y_1,\ldots,Y_n}(y_1,\ldots,y_n;\hat{\theta})} \\ &= \frac{p_C(1;\hat{s}) \prod_{i=1}^{n} p_{Y_n|C}(y_i|1;\hat{p}_i)}{p_C(1;\hat{s}) \prod_{i=1}^{n} p_{Y_i|C}(y_i|1;\hat{p}_i) + p_C(0;\hat{s}) \prod_{i=1}^{n} p_{Y_i|C}(y_i|0;\hat{q}_i)} \\ &= \frac{1}{1 + \dfrac{p_C(0;\hat{s})}{p_C(1;\hat{s})} \displaystyle\prod_{i=1}^{n} \frac{p_{Y_i|C}(y_i|0;\hat{q}_i)}{p_{Y_i|C}(y_i|1;\hat{p}_i)}}.\end{aligned}$$

(ii) Suppose a word appears in the new email (e.g., $Y_1 = 1$) but did not appear in the training data (i.e., $Y_1^{(j)} = 0$ for $j = 1, \ldots, k$). Compute $p_{Y_1,\ldots,Y_n}(y_1, \ldots, y_n)$. What went wrong?

Solution: We would have estimated $\hat{p}_1 = \hat{q}_1 = 0$, and

$$\begin{aligned} p_{Y_1,\ldots,Y_n}(y_1, \ldots, y_n; \hat{\theta}) &= p_C(1; \hat{s}) \prod_{i=1}^{n} p_{Y_i|C}(y_i|1; \hat{p}_i) + p_C(0; \hat{s}) \prod_{i=1}^{n} p_{Y_i|C}(y_i|0; \hat{q}_i) \\ &= p_C(1; \hat{s})\, p_{Y_1|C}(1|1; \hat{p}_1) \prod_{i=2}^{n} p_{Y_i|C}(y_i|1; \hat{p}_i) \\ &\quad + p_C(0; \hat{s})\, p_{Y_1|C}(1|0; \hat{q}_1) \prod_{i=2}^{n} p_{Y_i|C}(y_i|0; \hat{q}_i) \\ &= 0. \end{aligned}$$

Evidently, this observation is impossible under the learned model! Hence computing $\mathbb{P}(C = 1|Y_1 = 1, Y_2 = y_2, \ldots, Y_n = y_n)$ doesn't make sense since we're conditioning on an event that has 0 probability.

(iii) The result of part (ii) suggests that ML estimates can lead to a model that is fragile. Suggest a way to fix this problem.

Solution: There are many ways to do this. Here (and in the computational lab), we will employ *Laplace smoothing*, also called *additive smoothing*. This approach adds 1 to both the numerator and the denominator of the estimate, i.e., the estimates in part (a)(ii) are replaced with

$$\begin{aligned} \hat{p}_i' &= \frac{1}{2\alpha + \sum_{j=1}^{k} c^{(j)}} \left[\alpha + \sum_{j=1}^{k} c^{(j)}\, y_i^{(j)} \right], \quad i = 1, \ldots, n, \\ \hat{q}_i' &= \frac{1}{2\alpha + \sum_{j=1}^{k} 1 - c^{(j)}} \left[\alpha + \sum_{j=1}^{k} (1 - c^{(j)})\, y_i^{(j)} \right], \quad i = 1, \ldots, n, \end{aligned}$$

where α is a small value, say 1. The interpretation is that an empirical count of the number of times feature i appears in the spam (or not spam) emails is increased by *pseudo-count* α, and the denominator is adjusted appropriately to account for 2α increase (α for spam and α for not spam).